AI-Powered Multimodal Misinformation Detection and Verification System

Final Project Presentation

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Background

Misinformation spreads quickly online.
AI-generated and manipulated media can make false claims convincing.

Scope

Implemented: text, text-based PDF and image analysis.

Planned: video, OCR, SHAP and database

Objectives

1. Analyse text, PDF and image inputs.
2. Fine-tune DistilBERT and EfficientNet-B0.
3. Fuse text and image features using an MLP.
4. Return Real/Fake predictions with confidence.

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Existing Limitations

- Most tools focus on one media type.
- Text and image evidence are often analysed separately.
- Users need multiple tools.

Problem Addressed

Develop one lightweight web application that analyses supported inputs and combines text-image features to predict Real/Fake with confidence.

Literature Review – Selected Studies (1/2)

Paper / publication (year)	Method	Key contribution
Choras et al. (2021) *Applied Soft Computing*	Systematic mapping of ML methods	Summarises approaches, datasets and research challenges.
Abbas & Taeihagh (2025) *Int. J. Machine Learning and Cybernetics*	Multi-level semantic feature fusion	Combines semantic representations for multimodal classification.
Zhu et al. (2024) *Complex & Intelligent Systems*	Intra-modal aggregation and inter-modal semantic fusion	Models both within-modality features and text-image relationships.

Literature Review – Selected Studies (2/2)

Paper / publication (year)	Method	Key contribution
Nadeem et al. (2023) *J. King Saud Univ. – Computer and Information Sciences*	Stylometric and semantic similarity (SSM)	Combines writing-style, text-image similarity and image-forgery cues.
Yan et al. (2024) *J. King Saud Univ. – Computer and Information Sciences*	CLIP contrastive learning (SARD)	Aligns text-image semantics and incorporates social context.
Wang et al. (2024) *PLOS ONE*	LLM, CLIP and co-attention	Combines text, image-tampering and cross-modal signals with LLM support.

Literature Review – Comparative View

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Study	Year	Input	Main idea
Choras et al.	2021	Review	Maps ML-based fake-news detection research.
Nadeem et al.	2023	Text + image	Stylometric and semantic similarity.
Zhu et al.	2024	Text + image	Feature aggregation and semantic fusion.
Yan et al.	2024	Text + image + context	CLIP-based multimodal alignment.
Wang et al.	2024	Text + image	LLM-enhanced multimodal classification.
Abbas & Taeihagh	2025	Text + image	Multi-level semantic feature fusion.

Published results use different datasets and evaluation settings; they are not directly comparable.

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Research Gap

What prior work shows

Text-image models can improve detection, but often use complex architectures and different evaluation settings.

Our focus

A lightweight, integrated web application for text, text-based PDF and image analysis, with compact models and feature fusion.

Video, OCR, explainability, database storage and external fact verification are future extensions.

Proposed Solution

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Core Pipeline

1. Accept text, PDF or image.
2. Preprocess and extract features.
3. DistilBERT: 768-D text features.
4. EfficientNet-B0: 1280-D image features.
5. MLP fusion predicts Real/Fake and confidence

Technology

React + FastAPI

PyTorch + Transformers

PyMuPDF

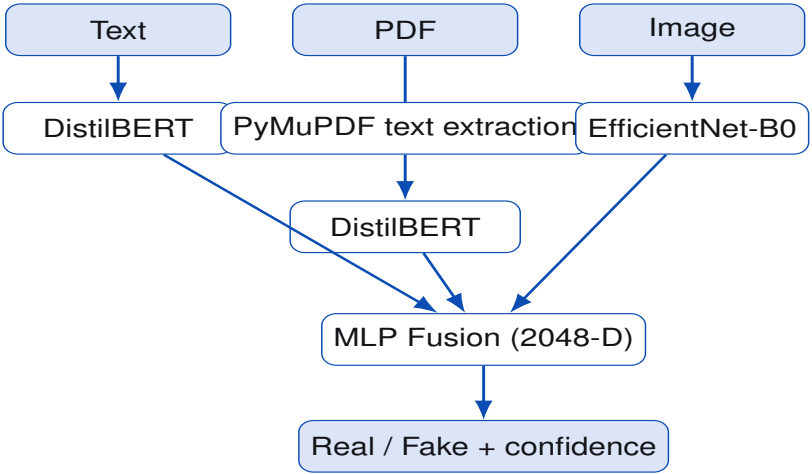
Fakeddit dataset

Design Goal

Compact, modular inference on a 4 GB GPU.

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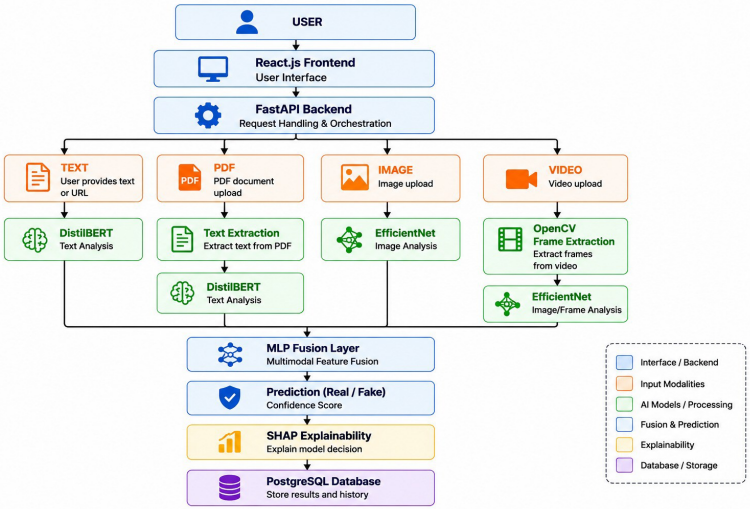
Proposed Solution – Input Handling



Video processing is planned and is not included in the current prediction pipeline.

System Architecture

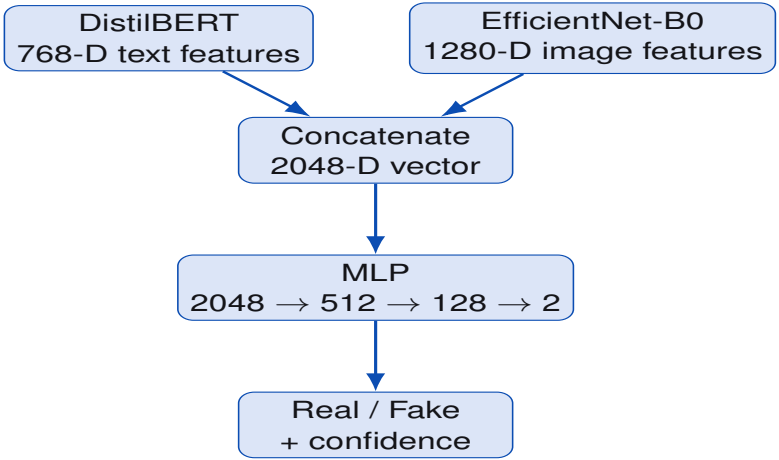
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Note:

video processing, SHAP explainability and PostgreSQL storage are planned extensions

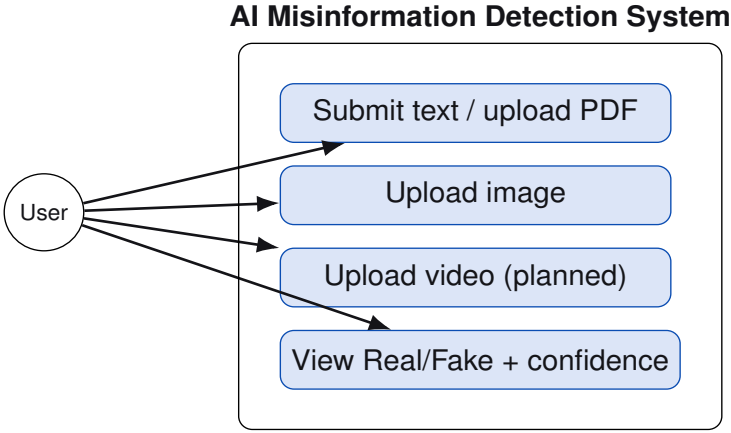
System Architecture – ML Pipeline



Feature-level early fusion: text and image representations are concatenated before classification.

Use Case Diagram

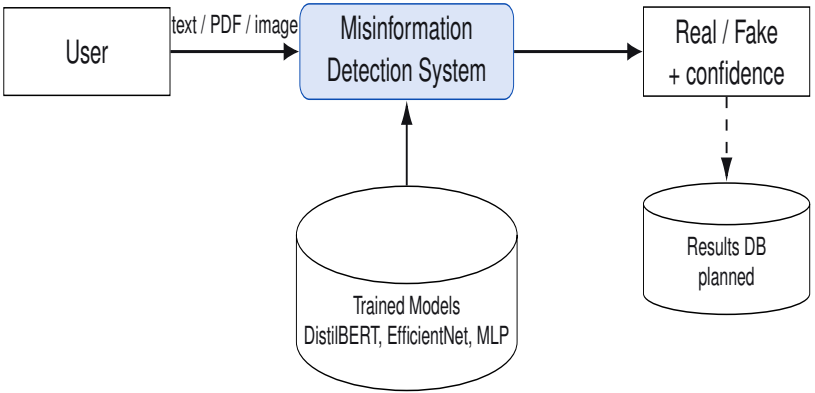
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Actor: User only. FastAPI and the ML models are internal system components, not external actors.

DFD Level 0 – Context Diagram

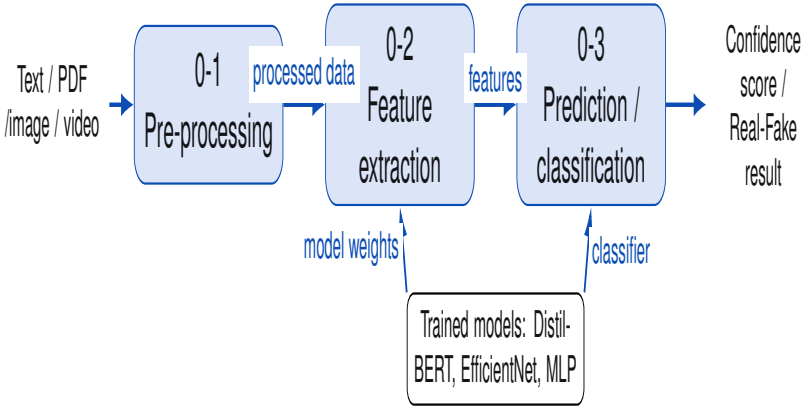
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Level 0 shows the system as one process, its external user, model store and output/result store.

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DFD Level 1 – Detailed Process Flow



Level 1 shows the internal processing stages only; no external actor is included.

Methodology

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Data and Preprocessing

- Fakeddit; labels 0 = Fake/Misinformation, 1 = True/Real.
- Text tokenized to max 128 tokens.
- Images resized to 224×224 and ImageNet-normalized.
- PDF text extracted using PyMuPDF.

Text Model

DistilBERT fine-tuned on a 50,000-sample training subset. Best observed validation accuracy: **86.36%**; F1: **82.58%**.

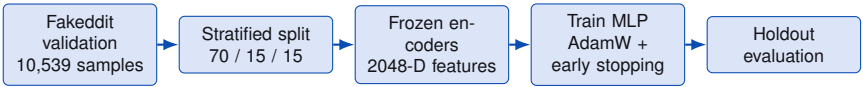
Image Model

EfficientNet-B0 transfer-learned on balanced 10K image subsets. Best new-10K experiment: **78.89% accuracy**, **80.00% F1**.

Fusion Model

Frozen DistilBERT and EfficientNet encoders produce 768-D and 1280-D features. Concatenation gives 2048-D input to an MLP: **2048** $\rightarrow$ **512** $\rightarrow$ **128** $\rightarrow$ **2**.

Fusion Training and Evaluation



Dataset split

7,377 training | 1,581 validation | 1,581 holdout

Evaluation note

The holdout was not used to train the MLP. Since encoders used this validation source during development, final testing should use the separate public Fakeddit test set.

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Implemented

- Accept text input.
- Upload and analyse text-based PDFs.
- Extract PDF text using PyMuPDF.
- Upload and analyse images.
- Text-only, image-only and text+image prediction.
- Display Real/Fake label and confidence.
- FastAPI prediction endpoint.

Planned / Extended

- Video upload with OpenCV frame extraction.
- SHAP explanations.
- PostgreSQL result history.
- External fact verification using trusted sources.
- OCR for scanned/image-only PDFs.

Non-functional Requirements

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Performance

- Compact pretrained models for practical inference.
- GPU-accelerated training.
- Early stopping.

Usability

- Single interface for supported input types.
- Clear prediction and confidence display.
- Simple upload/remove controls.

Reliability and Maintainability

- Modular frontend, backend and ML components.
- Input validation and error handling.
- Reproducible checkpoints and experiment results.

Scalability and Security

- REST API permits independent deployment.
- CORS configured for local frontend.
- Future database integration for controlled persistence.

Hardware Requirements

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Component	Requirement
Processor	Intel Core i5 / AMD Ryzen 5 or better
RAM	8 GB minimum; 16 GB recommended
Storage	256 GB SSD minimum; 512 GB recommended
GPU	NVIDIA GPU optional; experiments trained on RTX 2050 with 4 GB VRAM
Display / Client	Laptop or desktop with a modern web browser
Network	Internet useful for setup; local inference can run on localhost

Development Hardware

The project was developed and tested using an NVIDIA GeForce RTX 2050 laptop GPU with 4 GB VRAM.

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Software Requirements

Development Environment

- Windows 10/11 or Linux
- Python 3.10 environment
- Git and GitHub
- VS Code / terminal development
- Conda environment

Frontend / Backend

React.js, Vite, FastAPI, Uvicorn and REST API communication.

AI / Data Libraries

- PyTorch
- Hugging Face Transformers
- torchvision / timm
- OpenCV
- pandas, NumPy, scikit-learn
- PyMuPDF

Planned Components

PostgreSQL for persistent storage and SHAP for explainability.

Model Performance Results

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Model	Accuracy	Precision	Recall	F1
DistilBERT (50K)	86.36%	82.87%	82.29%	82.58%
EfficientNet-B0 (new 10K)	78.89%	73.43%	87.86%	80.00%
MLP Fusion – validation	88.17%	85.50%	90.79%	88.07%
MLP Fusion – holdout	**88.05%**	**85.64%**	**90.26%**	**87.89%**

Best observed result

The multimodal fusion model achieved **88.05% holdout accuracy** and **87.89% holdout F1**; early stopping selected the model with the highest validation F1.

Individual encoder and fusion experiments use different validation setups; the numbers are not a strict apples-to-apples benchmark.

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Fusion Results / API

Fusion Holdout Confusion Matrix

	Pred. Fake	Pred. Real
Actual Fake	706	115
Actual Real	74	686

Correct: 1,392 / 1,581
Accuracy: 88.05%

FastAPI PDF Test

Text-based PDF processed successfully.

Mode	PDF
Prediction	True / Real
Confidence	54.44%
Fake probability	45.56%
True probability	54.44%
Extracted text	1005 characters

End-to-end status

React + FastAPI prediction works for text, PDF and image inputs.

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Conclusion

- Developed a unified web-based framework for misinformation classification across text, PDF and image inputs.
- Fine-tuned DistilBERT and transfer-learned EfficientNet-B0 using Fakeddit data.
- Implemented feature-level multimodal fusion using a 2048-D representation and an MLP classifier.
- Achieved **88.05% accuracy and 87.89% F1** on the fusion holdout set used in the experiment.
- Integrated React with FastAPI and verified PDF-to-text inference using PyMuPDF.
- The modular design provides a foundation for future video analysis, explainability and external verification.

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Model and Modality Extensions

- Video analysis with frame extraction and temporal aggregation.
- OCR for scanned PDFs.
- Cross-modal attention for stronger fusion.
- Independent evaluation on Fakeddit test data.
- Multilingual misinformation detection.

Explainability and Verification

- SHAP explanations.
- Trusted-source retrieval for fact verification.
- PostgreSQL prediction history.
- Confidence calibration and uncertainty estimation.
- Larger concurrent deployment.

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